Transcript-Aware Mutation-Centred DNA Reconstruction for Lung Adenocarcinoma Stage Stratification: A Multi-Cohort Deep Learning and Federated Learning Study

Zohaib Alam[a], Didar Murad[a,b,*], Andrew Ware[b]

[a] *Computational Healthcare Intelligence (CHI) Lab, International Council for Research & Innovation in STE (ICRI-STE), Islamabad, Pakistan*
[b] *Faculty of Computing, Engineering, and Science, University of South Wales, Cardiff, UK*

Abstract

Lung adenocarcinoma (LUAD) shows substantial genomic variation between patients, making stage stratification from mutation-derived DNA sequences difficult. We developed a deep learning framework for LUAD stage stratification using both individual Stage I–IV prediction and Early (Stages I–II) versus Advanced (Stages III–IV) classification. Because four-stage prediction showed limited performance, the binary formulation was carried forward to the federated learning analysis. Mutation and clinical records from seven LUAD cohorts were combined with 96 LUAD-related genes identified from IntOGen, and transcript-specific coding sequences were obtained from Ensembl. After mutation reconstruction and quality checks, 89 genes retained at least one valid mutation representation. Initial experiments with 101-bp windows gave a CNN ROC-AUC of 0.6832. We later corrected sequence construction using transcript-specific coordinates, reference allele checks, strand information, and 301-bp mutation-centred windows, which improved ROC-AUC to 0.7652 for CNN, 0.7748 for LSTM, and 0.7769 for BiLSTM, the highest. The LSTM produced the highest binary accuracy (69.29%), with 79.55% recall for the Advanced group and an F1 score of 0.6809. Four-stage prediction remained harder. For the federated experiment, the BiLSTM was

*Corresponding author
Email address: didarmurad2019@gmail.com (Didar Murad)

trained across three simulated hospitals using weighted FedAvg for three communication rounds, with one local epoch per client per round. On the held-out test set of 712 patients, the matched centralized BiLSTM achieved accuracy, F1 score, and ROC-AUC of 0.7022, 0.7072, and 0.7808, compared with 0.6067, 0.6226, and 0.6756 for the federated model. The ROC-AUC difference of 0.1052 was statistically significant under the correlated DeLong test ($p = 8.64 \times 10^{-11}$). Federated validation ROC-AUC rose from 0.6432 to 0.6853 across three communication rounds. Overall, the results support mutation-centred genomic sequences for LUAD stage stratification and show that training across separate sites is possible without pooling patient-level training records. The system remains a research prototype without formal privacy guarantees or clinical diagnostic validation.

Keywords: Lung adenocarcinoma, Mutation-centred genomic DNA, Deep learning, BiLSTM, Federated learning, FedAvg

1. Introduction

Lung cancer continues to impose a major global health burden, with an estimated 2.5 million new cases and 1.8 million deaths reported in 2022 [1]. Non-small cell lung cancer (NSCLC) comprises several biologically distinct tumour types, among which lung adenocarcinoma (LUAD) is one of the most common [2]. Clinical stage is central to patient assessment because it reflects disease extent and contributes to prognosis and treatment planning [3]. However, LUAD is molecularly heterogeneous, and patients with similar clinical stages may differ substantially in their genomic alterations and disease characteristics [4]. This study investigates whether mutation-centred coding sequence information can support patient-level LUAD stage stratification.

Large-scale genomic studies have demonstrated the complexity of the LUAD mutational landscape. The Cancer Genome Atlas identified recurrent alterations in genes including TP53, KRAS, EGFR, STK11, KEAP1, and NF1 [4]. Individual tumours may contain multiple mutations affecting different genes, transcripts, and mutation classes, making patient-level representation important when modelling genomic disease patterns [4]. The datasets considered in the present study contain patients already diagnosed with LUAD and do not include an appropriate healthy or non-cancer control population. Consequently, the framework evaluates stage stratification

among LUAD patients rather than cancer detection. Both individual Stage I–IV prediction and Early (Stages I–II) versus Advanced (Stages III–IV) classification were examined.

Conventional machine learning approaches provide useful baselines for genomic prediction by representing mutations through structured variables such as gene identity, mutation counts, mutation classes, and clinical features. Although such representations are computationally efficient, they discard much of the nucleotide context surrounding individual variants. Sequence-based modelling offers a complementary approach in which local DNA patterns can be learned directly. Convolutional neural networks (CNNs), for example, have demonstrated the ability to identify informative sequence motifs from genomic data [5]. LSTM networks are designed to capture information that extends across longer parts of a sequence [6]. More recently, genomic Transformer models such as DNABERT have demonstrated contextual learning directly from nucleotide sequences [7]. The present study focuses on CNN, LSTM, and BiLSTM architectures because they provide practical sequence learning models for the available dataset and computational setting.

The usefulness of direct sequence modelling depends strongly on the biological consistency of the sequence representation. Coding mutations are transcript-specific, and substitutions, deletions, insertions, duplications, and deletion-insertion variants modify coding sequences in different ways. Incorrect transcript selection, strand orientation, reference allele interpretation, or coordinate handling can therefore introduce errors before model training. Standardized variant descriptions such as the Human Genome Variation Society (HGVS) nomenclature provide a framework for representing sequence variants consistently [8]. Reliable mutation-centred modelling should therefore combine computational sequence construction with transcript and reference allele validation rather than assuming that every reconstructed window is biologically valid.

The study developed through two sequence-reconstruction stages followed by a federated learning experiment. A preliminary 101-bp mutation-centred representation was first evaluated using conventional machine learning baselines and sequence-based modelling. Limitations identified in the reconstruction of complex mutation events motivated a corrected 301-bp pipeline incorporating transcript-specific coordinate validation, reference allele verification, strand-aware processing, and operation-specific mutation handling. The corrected representation was subsequently evaluated using CNN, LSTM, and BiLSTM architectures. This progression allowed the effect of a more carefully

validated mutation-centred sequence representation to be examined without repeating detailed reconstruction procedures outside the Methods section.

The corrected dataset integrated somatic mutation and clinical information from seven LUAD cohorts available through cBioPortal [9, 10], a study set of 96 LUAD-associated genes derived from IntOGen [11], and transcript-specific coding DNA sequences obtained from Ensembl [12]. Following the complete reconstruction and validation pipeline, 89 of the 96 genes retained at least one valid mutation representation. Validated mutation windows belonging to the same individual were combined into patient-level representations so that mutation-derived genomic information and selected clinical covariates could be modelled jointly.

Patient-level data separation and leakage-aware feature selection were used throughout the study. Cohort identity was excluded as a predictive input because the contributing studies differed in recruitment characteristics and stage distributions. Tumour-node-metastasis-derived variables were also excluded because they contribute directly to clinical stage assignment and could introduce target leakage. These choices were intended to ensure that model performance reflected mutation-derived genomic information and the permitted clinical covariates rather than direct encoding of stage.

Both individual Stage I–IV prediction and Early versus Advanced classification were evaluated for LUAD stage stratification. Exact four-class staging proved more difficult, particularly for the intermediate stages, and the binary Early/Advanced formulation was therefore carried forward to the federated learning analysis. The four-class results were retained as an exploratory evaluation of whether mutation-centred genomic representations could distinguish more specific disease stages.

A further challenge arises when genomic data are distributed across institutions. Centralized pooling of patient-level genomic records can create practical concerns involving data governance, ownership, institutional policies, and privacy [13, 14]. Federated learning provides an alternative framework in which clients train local model copies while the underlying training records remain decentralized. Federated Averaging (FedAvg) aggregates locally updated model parameters to form a shared global model [15]. Federated approaches have consequently received increasing attention in medical and healthcare applications [13, 14]. Importantly, federated learning reduces the need to pool raw records but does not by itself provide a formal privacy guarantee [14, 16].

In the final experimental phase, the binary BiLSTM was evaluated in a

simulated three-client federated environment using FedAvg and compared with a matched centralized BiLSTM on the same held-out test population. The federated experiment was designed to assess whether stage-related genomic signal could be learned while patient-level training records remained distributed across clients. Because differential privacy, secure aggregation, and other formal privacy mechanisms were not implemented, the system is treated as a research prototype rather than a formally privacy-preserving clinical framework.

Overall, this study contributes a transcript- and reference-validated 301-bp mutation-centred sequence pipeline, patient-level CNN, LSTM, and BiLSTM modelling, leakage-aware evaluation of binary and four-class LUAD staging, and a matched centralized versus federated BiLSTM comparison. In contrast to approaches based primarily on engineered mutation features or unvalidated sequence windows, the present framework reconstructs patient-level mutation-centred DNA representations using transcript-specific coordinates, reference allele verification, strand-aware processing, and operation-specific mutation handling before model training. The corrected representation is evaluated under both centralized and federated settings using matched patient populations, allowing decentralized optimisation to be examined without changing the underlying genomic representation. The work is intended to assess the feasibility of mutation-aware genomic stage stratification and decentralized model training rather than cancer diagnosis or clinical deployment.

2. Materials and Methods

The overall study workflow is summarized in Fig. 1.



Figure 1: Overall workflow of the LUAD stage-stratification study. Mutation and clinical data from seven cBioPortal cohorts were integrated with IntOGen-derived LUAD genes and transcript-specific Ensembl coding sequences. The preliminary 101-bp pipeline was followed by the corrected 301-bp mutation-centred reconstruction and patient-level representation. Labelled patients were then used for centralized and federated model development and evaluation.

2.1. Data Sources and Study Cohorts

Somatic mutation and clinical data for lung adenocarcinoma (LUAD) were obtained from seven publicly available cohorts through cBioPortal: TCGA PanCancer Atlas 2018 [1], MSK 2017 [2], MSKCC 2020 [3], MSKCC 2023 [4], MSK NPJ Precision Oncology 2021 [5], NCI 2022 [6], and OncoSG

2020 [7]. All seven cohorts were accessed through cBioPortal [8, 9]. Multiple cohorts were included to increase patient diversity and support patient-level stage stratification across independently generated LUAD datasets.

Ninety-six LUAD-associated genes were extracted from IntOGen LUAD mutation and driver gene data and used as the initial study gene set [10]. Transcript-specific coding DNA sequences were obtained from Ensembl and used as reference sequences for mutation reconstruction [11].

The preliminary 101-bp and corrected 301-bp mutation-centred pipelines were processed separately and therefore produced distinct modelling populations. The final corrected 301-bp patient-level dataset contained 4,998 patients, of whom 3,558 had usable stage labels for supervised modelling. Because no appropriate healthy or non-cancer controls were available, the study evaluates LUAD stage stratification rather than cancer diagnosis.

2.2. Clinical Data Processing and Stage Definition

Clinical records from the seven LUAD cohorts were harmonized before modelling. Pathological stage labels were standardized as Stage I, Stage II, Stage III, or Stage IV. For binary classification, Stages I–II were grouped as Early and Stages III–IV as Advanced [1]. Among the 4,998 patients in the corrected 301-bp dataset, 3,558 had usable stage labels: 1,627 Stage I, 465 Stage II, 590 Stage III, and 876 Stage IV, corresponding to 2,092 Early and 1,466 Advanced patients. The remaining 1,440 patients were retained in the integrated dataset but excluded from supervised modelling.

Clinical covariates comprised sex, age, active smoking status, and pack-years. Smoking status was encoded as never smoker (0) versus ever smoker (1), including former, current, or otherwise documented ever smokers. Unknown smoking status was retained as missing during harmonization. Of the 4,998 patients, 95 had missing active smoking values, all of whom belonged to the unlabelled group; therefore, no smoking status imputation was required for supervised modelling.

Age and pack-years were converted to numeric form. For centralized modelling, missing continuous values were imputed using medians estimated from the training partition only, while categorical variables used training-derived modes when required. Continuous clinical variables were standardized using parameters fitted on the centralized training data and then applied unchanged to the corresponding validation and test sets. For the federated experiment, shared preprocessing parameters were fitted using the 2,846-patient federated development pool before the local train and validation par-

titions were formed. Cohort identity, stage labels, and tumour, node, and metastasis-derived variables were excluded from predictive inputs to avoid direct or indirect target leakage.

2.3. Preliminary 101-bp Mutation-Centred Representation

The preliminary 101-bp pipeline initially produced 15,142 mutation rows. After sequence quality filtering, 12,689 rows remained technically valid, and 12,580 were retained after representative sample alignment. All 3,382 patients had at least one retained mutation from the selected sample. The mean number of mutation windows was 3.72 per patient, with a maximum of 37.

Each mutation was represented by paired reference and mutated 101-bp DNA windows. A mutation position mask identified affected positions, while a padding mask distinguished biological sequence from boundary padding. Mutation consequence severity was retained only as a computational feature and was not interpreted as clinical risk. The patient-level representation included 96 gene indicators, of which 89 were active in the final dataset. Because smoking status and pack-years were inconsistently available across cohorts, they were excluded from this preliminary experiment. Numerical clinical features comprised encoded sex, scaled age, an age unusable indicator, and scaled mutation count.

Patients were split at the patient level into training, validation, and test sets using a 70/15/15 ratio, yielding 2,364, 507, and 511 patients, respectively, with no overlap between partitions. For sequence modelling, at most 13 mutation windows were retained per patient as a fixed modelling cap; this value did not represent the maximum mutation burden in the dataset, which reached 37 mutations per patient before truncation. Before truncation, mutations were ordered by decreasing severity, followed by affected span length and stable original order.

The mask-aware CNN therefore received seven inputs: reference DNA windows, mutated DNA windows, mutation position masks, mutation severity, window padding masks, 89 active gene indicators, and four numerical patient-level features. This subsection describes only the preliminary 101-bp representation; the corrected 301-bp reconstruction is described separately below.

The conventional machine learning experiments were retained as preliminary baselines, whereas subsequent modelling focused on deep sequence

learning architectures because the primary study objective was to learn directly from mutation-centred nucleotide windows rather than rely solely on engineered patient-level features.

2.4. Transcript and Coding Sequence Validation

For the corrected 301-bp reconstruction, transcript-specific coding DNA sequences for the 96-gene study set were obtained from Ensembl [2]. The FASTA collection contained 2,795 transcript records, of which 579 were non-sequence placeholders, leaving 2,216 usable coding sequences. All 96 genes retained at least one valid coding sequence.

After representative sample selection, 21,239 mutation rows were evaluated for transcript matching. Transcript identifiers were available for 21,236 mutations, and 20,184 matched an exact usable Ensembl transcript. The remaining 1,055 mutations were excluded: 1,052 referred to transcripts absent from the valid FASTA set and three lacked a transcript identifier. Mutations were not reassigned to alternative transcripts because coding level HGVS coordinates are transcript-specific [3].

HGVSc notation was parsed to identify substitutions, deletions, duplications, deletion-insertion variants, and insertions. Operations were successfully resolved for 19,478 mutations, and 18,534 retained valid coding sequence coordinates after coordinate checking. Reference alleles were then verified against the matched Ensembl coding sequence, with insertions assessed using flanking sequence validation. This yielded 17,992 reference-verified mutation payloads; 542 reference-mismatched mutations were excluded before 301-bp window generation.

The corrected 301-bp dataset was reconstructed independently from the original cohort mutation files and Ensembl coding sequences; no sequence windows from the preliminary 101-bp pipeline were reused.

2.5. Corrected 301-bp Mutation Reconstruction

Fresh 301-bp reference windows were generated for all 17,992 reference-verified mutation payloads. Each window contained 150 bases upstream of the mutation anchor, one anchor position, and 150 bases downstream. For substitutions, the anchor was the mutated coding position; for deletions, duplications, and deletion-insertion variants, it was the midpoint of the affected CDS span, using the left-middle position for even-length spans; for insertions, the left flanking position was used. Sequence positions extending beyond CDS boundaries were padded only where unavailable using `N`.

Boundary padding was required for 2,658 reference windows (2,177 left and 481 right), with no window padded on both sides. All reference windows were exactly 301 bp long, and no mutation event extended beyond its assigned window.

Each mutation was applied to the matched full Ensembl coding sequence before extraction of the corresponding mutated 301-bp window. Operation-specific rules were used for substitutions, deletions, insertions, duplications, and deletion-insertion variants. All 17,992 mutations were applied successfully. Five deletion-insertion events produced sequences identical to their corresponding reference windows and were removed as no-op variants. The final reconstruction therefore contained 17,987 valid reference and mutated pairs: 14,856 substitutions, 1,978 deletions, 538 duplications, 518 deletion-insertion variants, and 97 insertions.

For each final pair, aligned 301-position binary mutation masks were generated for both the reference and mutated windows, with affected positions encoded as 1 and unaffected positions as 0 using operation-specific masking rules. All 17,987 pairs produced valid masks, with no mask format errors, affected position count mismatches, or overlap between mutation-positive mask positions and padded sequence positions.

2.6. Patient-Level Representation and Model Inputs

The 17,987 validated mutation reconstructions were aggregated into a patient-level dataset of 4,998 patients. Gene symbols, mutated 301-bp windows, and mutation-affected masks were stored as aligned variable-length lists. Among the 3,558 labelled patients, 3,459 had at least one valid reconstruction, whereas 99 had empty mutation lists. These patients were retained and represented using a dedicated `NO_MUTATION` entry.

The study initially extracted 96 LUAD-associated genes from the IntOGen-derived mutation and driver gene data, and all 96 genes were retained when patient-level gene indicator features were constructed. However, mutations were required to pass the complete corrected 301-bp reconstruction pipeline, including exact transcript matching, HGVSc operation resolution, coding coordinate validation, reference allele verification, successful operation-specific sequence reconstruction, and final sequence/mask validation. After these steps, valid mutation reconstructions represented 89 of the 96 genes. The remaining seven genes—*CAPN5*, *CBLB*, *CRLF2*, *EPPK1*, *NOTCH1*, *ZFHX4*, and *ZNF724*—had all-zero patient-level mutation indicators because no mutation in these genes remained as a valid final 301-bp reconstruction. These

genes were therefore not manually removed on the basis of biological relevance; rather, they had no valid mutation representation available for modelling after the complete validation pipeline. Thus, the 96 genes represent the initial IntOGen-derived study gene set, whereas 89 genes were actively represented by at least one valid reconstructed mutation in the final modelling data.

After this handling step, labelled patients contained a mean of 3.63 mutation entries, a median of 3, and a maximum of 37. The maximum across the full 4,998-patient dataset was 39. A fixed dimension of 37 mutation slots was therefore sufficient for all labelled patients, with no truncation.

The corrected 301-bp models used seven raw inputs: gene symbol lists, mutated DNA window lists, mutation-affected mask lists, sex, age, active smoking status, and pack-years. Reference windows, mutation severity features, and explicit sequence boundary padding masks used in the preliminary pipeline were not supplied as predictive inputs. Reference sequences were retained only for reconstruction and validation. DNA symbols were encoded as `PAD`, `A`, `C`, `G`, `T`, and `N`. For centralized modelling, the gene vocabulary and clinical preprocessing parameters were fitted using the training data only. For the federated experiment, shared preprocessing parameters were fitted using the 2,846-patient federated development pool before the local training and validation partitions were formed.

2.7. Dataset Splitting and Leakage Prevention

For the centralized 301-bp experiments, the 3,558 stage-labelled patients were stratified by the Early/Advanced target into 2,490 training, 534 validation, and 534 test patients. Early/Advanced counts were 1,464/1,026, 314/220, and 314/220, respectively, with no patient or sample overlap between partitions.

Clinical imputation, scaling, gene vocabulary construction, and class weights were derived from the training partition only. Cohort identity and target-related variables were excluded from predictive inputs. Validation ROC-AUC guided checkpointing and early stopping, while validation loss controlled learning rate reduction. The test set was not used for early stopping, hyperparameter selection, or threshold optimization; binary predictions used a fixed threshold of 0.50.

For the BiLSTM, the test set had been evaluated once before a later continuation training phase. Those results were not used for gradient updates,

checkpointing, early stopping, hyperparameter selection, or threshold selection; nevertheless, the test set was not strictly unseen before continuation.

2.8. Centralized Model Development

Three centralized models were evaluated on the corrected 301-bp representation: CNN, unidirectional LSTM, and BiLSTM. All used the same genomic and clinical inputs, preprocessing, patient partitions, and training-derived class weights.

The CNN used an eight-dimensional nucleotide embedding followed by Conv1D layers with 32 and 64 filters and kernel sizes of 7 and 5. The recurrent models used eight-dimensional DNA and gene embeddings, global sequence pooling, mutation-affected position pooling, and masked mean/max aggregation across mutations. The LSTM contained one 32-unit recurrent layer, whereas the BiLSTM used 32 units in each direction. Clinical features were processed separately and fused with the genomic representation before sigmoid classification. The CNN, LSTM, and BiLSTM contained 29,281, 22,305, and 31,777 trainable parameters, respectively. Inspection of the saved LSTM checkpoint confirmed a single recurrent weight set, verifying the unidirectional architecture.

All binary models used Adam with an initial learning rate of 10^{-3} [4], binary cross-entropy loss, and class weighting to account for unequal class frequencies [5]. The CNN was trained for up to 30 epochs with batch size 64, while the LSTM used 20 epochs with batch size 32 and completed the full run. Validation ROC-AUC guided checkpointing and early stopping, and validation loss controlled learning rate reduction.

The BiLSTM checkpoint was subsequently used for continuation training with batch size 32 and a maximum of 20 additional epochs. Validation guided early stopping terminated this phase after seven epochs. No corresponding continuation phase was applied to the CNN or LSTM. See the CNN and BiLSTM architectures in Fig. 2.

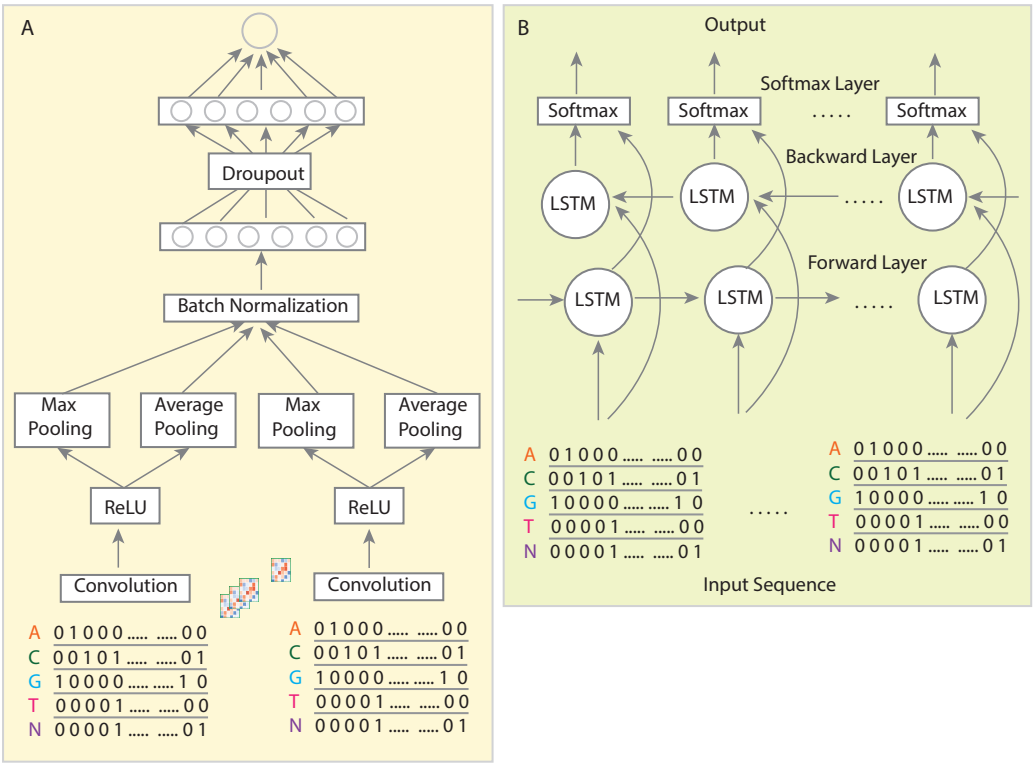


Figure 2: **Schematic architectures of the CNN and BiLSTM models.** (A) CNN feature extraction using shared convolutional filters, ReLU activation, max/average pooling, batch normalization, and dropout. (B) Bidirectional LSTM layers process the extracted sequence in forward and backward directions, followed by sigmoid classification to generate the final output prediction.

2.9. Federated Learning Design

The binary BiLSTM was extended to a simulated federated learning setting in which model training was distributed across multiple clients without centrally pooling raw patient records [6]. From the 3,558 labelled patients, a fixed stratified split with random seed 42 allocated 2,846 patients to the federated training pool and 712 patients to a held-out global test set. The training pool contained 1,673 Early and 1,173 Advanced cases, while the test set contained 419 Early and 293 Advanced cases.

The federated training pool was deterministically partitioned by cohort and binary stage strata across three simulated clients containing 949, 949, and 948 patients. All seven source cohorts were represented in each client. An 80/20 stratified local split produced training sets of 759, 759, and 758

patients, respectively, with 190 validation patients per client. No patient overlapped across clients or with the global held-out test set.

Each client received an identical copy of the global BiLSTM and trained locally for one epoch per communication round using a batch size of 16 and locally calculated class weights. Only model parameters, local training sample counts, and evaluation metrics were returned to the central server; raw genomic and clinical patient records remained local. The server updated the global model using training sample-weighted Federated Averaging (FedAvg) [7]:

$$w^{(r+1)} = \frac{\sum_{k=1}^{K} n_k w_k^{(r+1)}}{\sum_{k=1}^{K} n_k},$$

where $K = 3$, n_k denotes the number of local training patients at client k, and $w_k^{(r+1)}$ represents the locally updated model parameters. Three communication rounds were performed, with the aggregated global model redistributed to all clients after each round.

For a controlled comparison with federated training, a matched centralized BiLSTM baseline was trained using the same 2,846-patient development population and evaluated on the identical 712-patient held-out global test set. The development population was partitioned into 2,276 training and 570 validation patients using the deterministic local stratification procedure used in the federated experiment. Centralized preprocessing parameters were fitted using the 2,276 training patients only. The matched centralized model used the same BiLSTM architecture, a batch size of 16, training-derived class weights, and three training epochs, allowing centralized and federated performance to be compared on the same test population.

After the third communication round, the final global BiLSTM was evaluated on the 712-patient held-out global test set for Early (Stages I–II) versus Advanced (Stages III–IV) classification. The federated architecture reduces the need to centralize raw patient records but should not be interpreted as providing formal privacy guarantees [8, 9]. Differential privacy, secure aggregation, and encrypted model update mechanisms were not implemented in the present research prototype. The federated learning workflow is illustrated in Fig. 3.

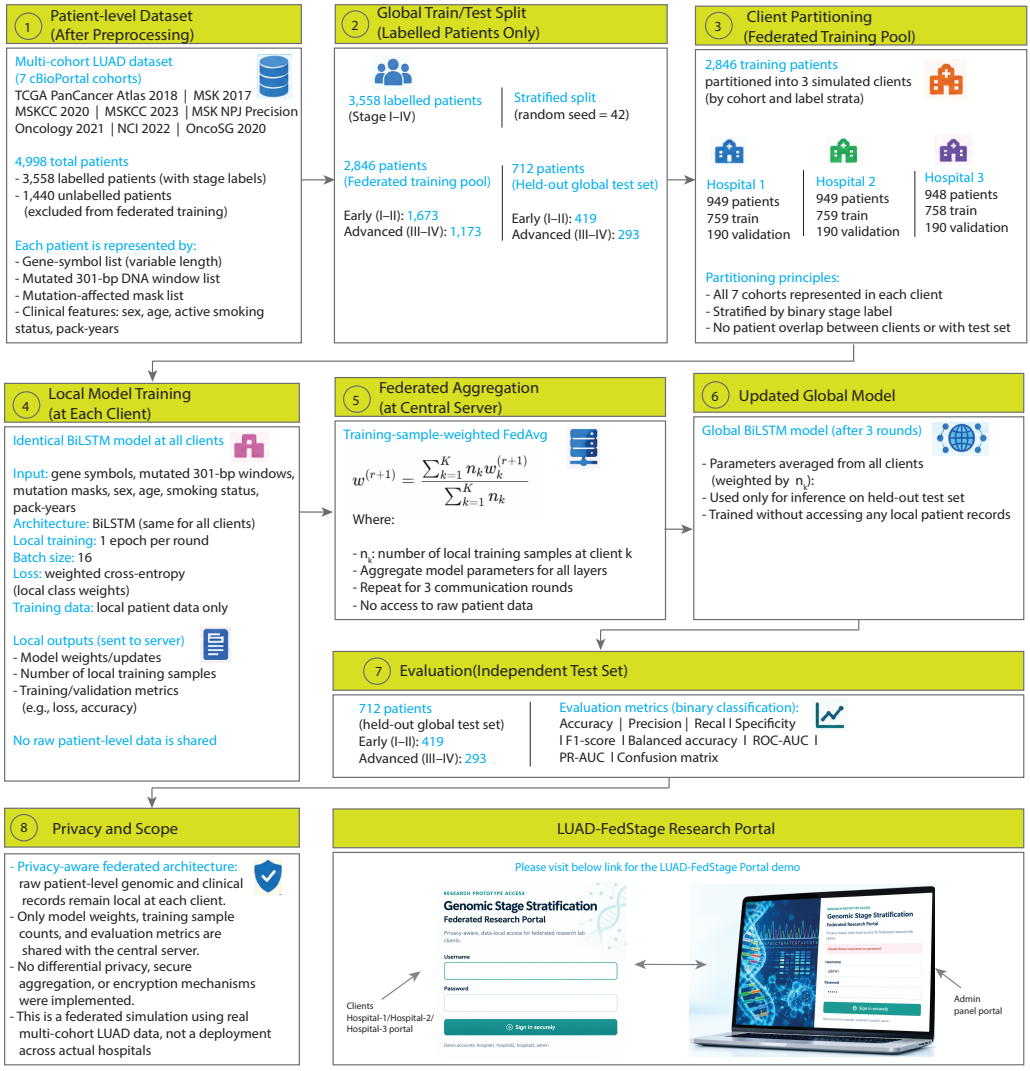


Figure 3: Federated learning workflow for binary LUAD stage stratification. The 3,558 labelled patients were divided into a 2,846-patient federated training pool and a 712-patient held-out global test set. The training pool was distributed across three simulated clients with local training and validation splits of 759/190, 759/190, and 758/190 patients. Identical BiLSTM models were trained locally for one epoch per round and aggregated using training-sample-weighted FedAvg over three communication rounds. Raw patient-level records remained local, and the final global model was evaluated on the held-out test set. **https://github.com/partnerships2024/Federated-Learning-LUAD-Stage-Stratification.git**

2.10. Model Evaluation

Binary model performance was evaluated using accuracy, precision, recall for the Advanced class (sensitivity), specificity, F1-score, balanced accuracy, ROC-AUC, PR-AUC, and confusion matrices. ROC-AUC summarized the models' ability to distinguish Early from Advanced disease across decision thresholds [10]. PR-AUC was also reported because the two classes were not equally represented [11]. Advanced disease was defined as the positive class and Early disease as the negative class, and binary predictions were obtained using a fixed threshold of 0.50.

For the matched centralized and federated BiLSTM comparison on the same 712-patient held-out test set, uncertainty was quantified using a stratified paired patient-level bootstrap with 10,000 resamples and random seed 42. Sampling was performed with replacement within the Early and Advanced classes, and 95% percentile confidence intervals were calculated for accuracy, F1-score, ROC-AUC, sensitivity, specificity, balanced accuracy, and PR-AUC. Paired bootstrap differences were calculated as centralized minus federated performance. Because both models were evaluated on the same patients, the difference in ROC-AUC was additionally assessed using the correlated DeLong test. All threshold-dependent binary metrics used the fixed decision threshold of 0.50.

The four-class Stage I–IV task was evaluated using accuracy, balanced accuracy, macro- and weighted precision, recall, and F1-score, together with top-2 accuracy and confusion matrices. Top-2 accuracy measured whether the correct stage appeared among the two highest-scoring classes. Class-balanced measures were retained because overall accuracy alone can provide an incomplete assessment under unequal class distributions [5]. PR-AUC was not used for the multiclass analysis.

Both the binary Early/Advanced and four-class Stage I–IV formulations were evaluated for LUAD stage stratification. Because the four-class models showed limited performance, the binary formulation was carried forward to the federated learning analysis, while the four-class results were retained as an exploratory evaluation.

3. Results

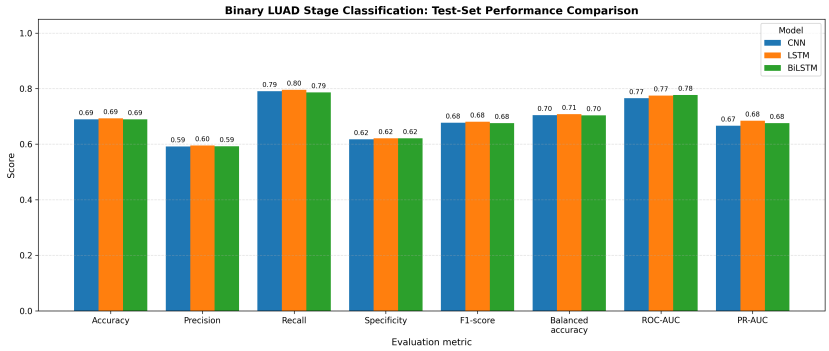
3.1. Binary Classification Performance

The three centralized models performed similarly on the held-out test set for binary Early/Advanced LUAD stage classification (Fig. 4(a)). Accu-

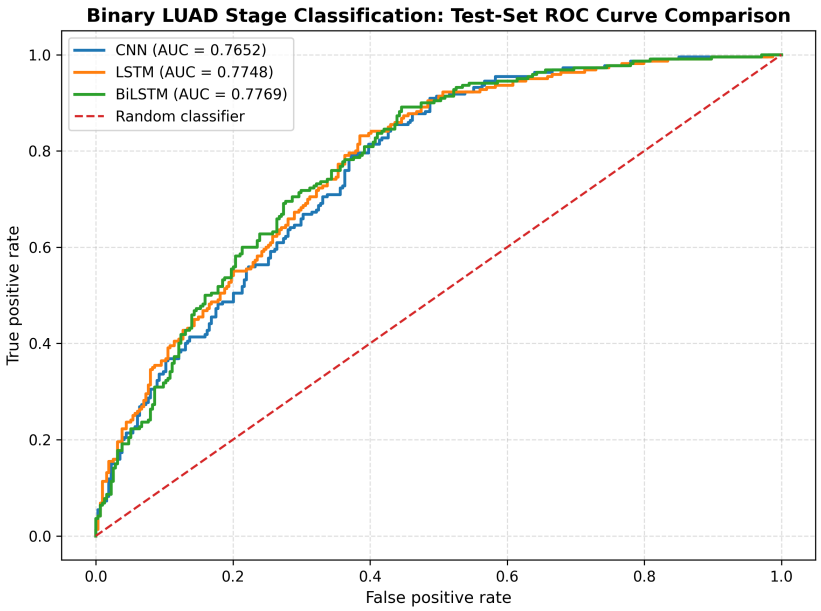
racy was 0.6891 for CNN, 0.6929 for LSTM, and 0.6891 for BiLSTM. The corresponding F1-scores were 0.6770, 0.6809, and 0.6758.

Recall for the Advanced group was higher than specificity for all three models. Recall was 0.7909 for CNN, 0.7955 for LSTM, and 0.7864 for BiLSTM, while specificity remained approximately 0.62 across the models. Precision ranged from 0.5918 to 0.5953. This shows that the stronger sensitivity to Advanced disease was accompanied by more Early-stage patients being classified as Advanced.

The LSTM produced the highest balanced accuracy, at approximately 0.708, and also had the highest Advanced-stage recall. The BiLSTM, however, produced the highest ROC-AUC. For ROC-AUC, the observed values were 0.7652 for CNN, 0.7748 for LSTM, and 0.7769 for BiLSTM. PR-AUC remained approximately 0.67–0.68. The differences between the models were small, and no architecture performed best across all metrics. No formal significance testing was performed for pairwise differences among these three centralized architectures, so these differences are reported descriptively. The centralized model evaluation is illustrated in Fig. 4.



(a) Held-out test-set performance comparison of the centralized CNN, LSTM, and BiLSTM models.



(b) Test-set ROC-curve comparison.

Figure 4: Evaluation of centralized models for binary Early/Advanced LUAD stage classification using the corrected 301-bp representation. (a) Held-out test-set performance comparison of the CNN, LSTM, and BiLSTM models. (b) Test-set ROC curves for the three centralized architectures.

3.2. Centralized and Federated BiLSTM Comparison

The matched centralized and federated BiLSTM models were evaluated on the same 712-patient held-out test set. The centralized model achieved an accuracy of 0.7022 (95% CI: 0.6699–0.7346), an F1-score of 0.7072 (95% CI:

0.6790–0.7345), and a ROC-AUC of 0.7808 (95% CI: 0.7466–0.8140). The corresponding federated results were 0.6067 (95% CI: 0.5716–0.6404), 0.6226 (95% CI: 0.5914–0.6521), and 0.6756 (95% CI: 0.6359–0.7148), respectively (Table 1).

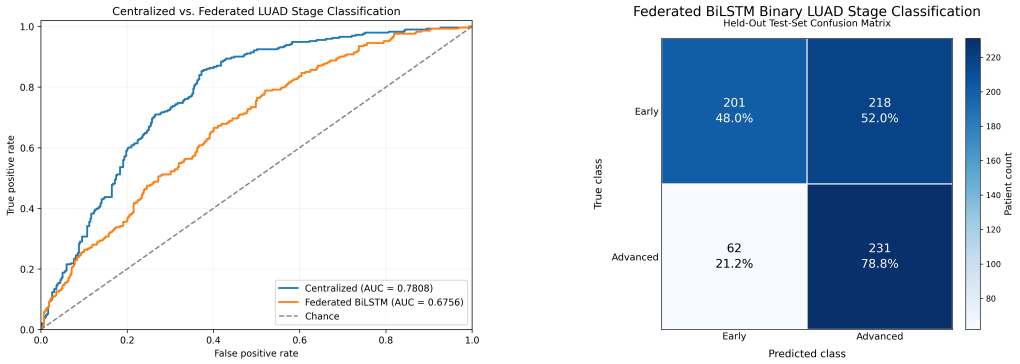
The paired differences favoured centralized training by 0.0955 in accuracy (95% CI: 0.0604–0.1306), 0.0845 in F1-score (95% CI: 0.0525–0.1178), and 0.1052 in ROC-AUC (95% CI: 0.0738–0.1375). The difference in ROC-AUC was also significant under the correlated DeLong test ($p = 8.64 \times 10^{-11}$). These results indicate stronger predictive performance under centralized training in the matched comparison. The federated model nevertheless retained discrimination above chance while patient-level training records remained distributed across three simulated clients. This finding should not be interpreted as evidence that federated learning improves predictive performance relative to centralized training. Federated validation performance continued to improve across the three communication rounds, as reported in the following subsection.

Table 1: Performance of matched centralized and federated BiLSTM models on the 712-patient held-out binary test set. Confidence intervals were estimated using 10,000 stratified paired patient-level bootstrap resamples.

Metric	**Centralized BiLSTM**	**Federated BiLSTM**	**Difference**
Accuracy	**0.7022** (0.6699–0.7346)	0.6067 (0.5716–0.6404)	0.0955 (0.0604–0.1306)
F1-score	**0.7072** (0.6790–0.7345)	0.6226 (0.5914–0.6521)	0.0845 (0.0525–0.1178)
ROC-AUC	**0.7808** (0.7466–0.8140)	0.6756 (0.6359–0.7148)	0.1052 (0.0738–0.1375)

Values in parentheses are 95% bootstrap confidence intervals. Differences are calculated as centralized minus federated performance. The paired ROC-AUC difference was significant using the correlated DeLong test ($p = 8.64 \times 10^{-11}$).

The discrimination performance and class-specific prediction pattern of the final federated BiLSTM are summarized in Fig. 5. The model achieved a ROC-AUC of 0.6756 (95% CI: 0.6359–0.7148) on the 712-patient held-out test set. Of the 419 Early-stage patients, 201 were classified as Early and 218 as Advanced. Of the 293 Advanced-stage patients, 231 were correctly classified as Advanced and 62 as Early.



(a) Held-out test-set ROC curve.

(b) Held-out test-set confusion matrix.

Figure 5: Evaluation of centralized and federated BiLSTM models on the matched 712-patient test set: (a) ROC curve comparison and (b) Confusion matrix with patient counts and row-wise percentages.

3.3. Federated Learning Dynamics

Federated validation performance improved across all three communication rounds. After each FedAvg step, the aggregated global model was evaluated separately on each simulated hospital's validation set, and round-level results were reported as validation-sample-count-weighted averages.

Weighted validation ROC-AUC increased from 0.6432 in round 1 to 0.6659 in round 2 and 0.6853 in round 3 (Fig. 6). F1-score increased from 0.5765 to 0.5924 and 0.6088, while accuracy rose from 0.5895 to 0.5930 and 0.6158. Validation loss decreased concurrently from 0.6938 to 0.6773 and 0.6626.

Because performance was still improving when training ended after three communication rounds, convergence of the federated optimisation process was not demonstrated. Additional rounds would be required to determine whether performance would stabilise or improve further.

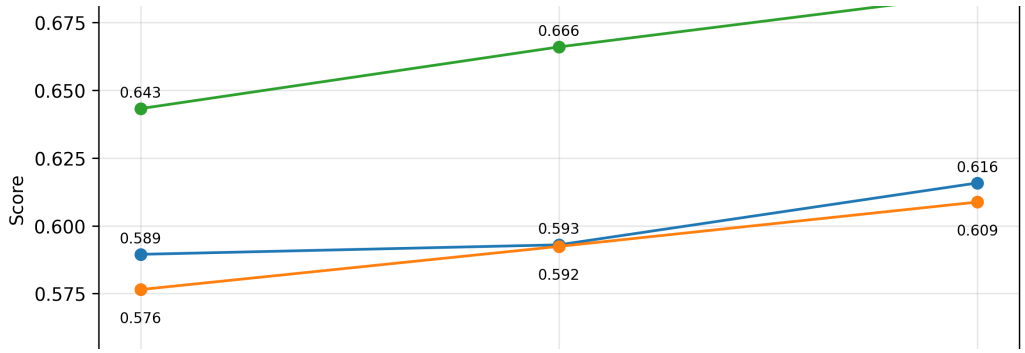


Figure 6: Round-wise validation performance of the federated BiLSTM across three communication rounds. Values are validation-sample-count-weighted averages obtained by evaluating the post-FedAvg global model separately on each hospital validation partition.

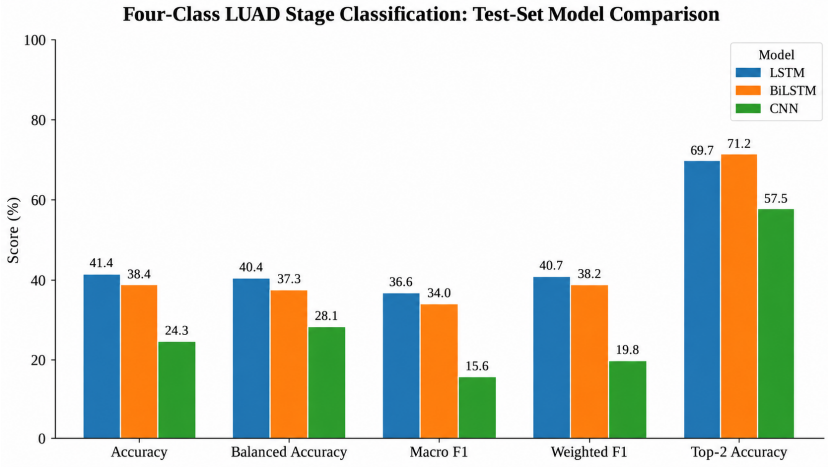
3.4. Four-Class Stage Classification

Four-class Stage I–IV prediction was evaluated using the centralized CNN, LSTM, and BiLSTM models. The 534-patient test set contained 242 Stage I cases (45.32%), so a majority-class classifier predicting Stage I for every patient would achieve an accuracy of 0.4532 and a balanced accuracy of 0.25.

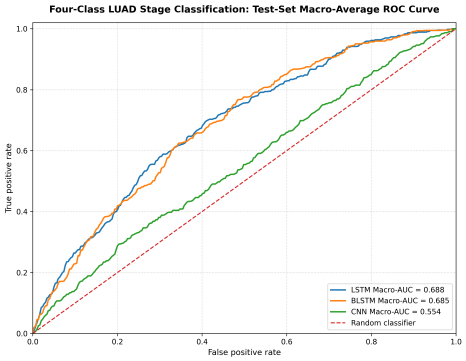
None of the three models exceeded the majority-class baseline in raw accuracy. Accuracy was 0.4139 for LSTM, 0.3839 for BiLSTM, and 0.2434 for CNN (Fig. 7(a)). When class balance was taken into account, the corresponding balanced accuracy values were 0.4038 for LSTM, 0.3732 for BiLSTM, and 0.2806 for CNN. These results indicate better separation across the four classes than would be obtained from the majority-only baseline.

The LSTM also achieved the highest macro F1-score (0.3657) and weighted F1-score (0.4072), compared with 0.3395 and 0.3820 for BiLSTM and 0.1560 and 0.1976 for CNN. Top-2 accuracy was highest for BiLSTM at 0.7116, followed by LSTM at 0.6966 and CNN at 0.5749.

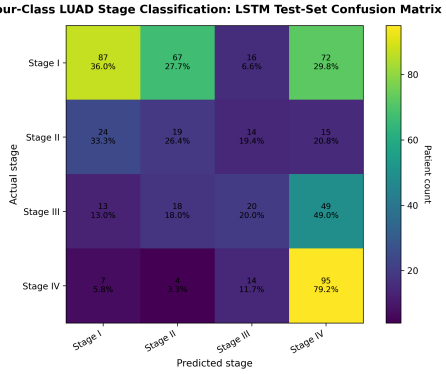
Overall, exact Stage I–IV classification remained difficult. The LSTM provided the strongest exact-stage performance, while the BiLSTM achieved the highest top-2 accuracy. Because no model exceeded the majority-class baseline in raw accuracy, the four-class results were retained to document the difficulty of exact stage prediction, while the better-performing binary formulation was carried forward to the federated learning analysis. The four-class stage classification results are summarized in Fig. 7.



(a) Centralized LSTM, BiLSTM, and CNN performance.



(b) Macro-average ROC curves.



(c) LSTM confusion matrix.

Figure 7: Evaluation of centralized models for four-class LUAD Stage I–IV classification using the corrected 301-bp representation. (a) Test-set performance comparison of the LSTM, BiLSTM, and CNN. (b) Test-set macro-average ROC curves. (c) LSTM confusion matrix showing patient counts and row-wise percentages for each true stage.

4. Discussion

4.1. Principal Findings

This study evaluated mutation-centred genomic representations for patient-level LUAD stage prediction using centralized and federated deep learning. Exact Stage I–IV classification was investigated as part of the main stage-prediction objective, but performance remained limited. The Early/Advanced binary formulation showed stronger discrimination and was therefore carried

forward to the federated learning analysis. Recent studies have continued to examine artificial intelligence for lung cancer characterisation and mutation-related prediction, including approaches focused on oncogenic alterations and multimodal disease representation [12, 13, 14].

The centralized CNN, LSTM, and BiLSTM showed similar binary performance, with accuracies of approximately 0.69 and ROC-AUC values of 0.77–0.78. This level of discrimination is moderate and is not sufficient to support standalone clinical staging. Advanced-stage sensitivity exceeded Early-stage specificity across all models, indicating stronger detection of Advanced disease at the cost of more Early-stage false positives. The small differences among architectures suggest that increased model complexity alone did not substantially improve discrimination under the present representation.

Performance improved after replacing the preliminary 101-bp representation with the corrected 301-bp mutation-centred reconstruction, with CNN ROC-AUC increasing from 0.6832 to 0.7652. Because several reconstruction improvements were introduced together, their individual contributions cannot be isolated from the present experiment. The findings nevertheless indicate that careful transcript-aware reconstruction and sequence validation are important when mutation-centred genomic windows are used for stage modelling. Mutation-centred genomic information should therefore be considered a complementary source of information rather than a replacement for imaging, pathology, TNM staging, or other clinical variables [15, 16].

In the matched comparison, centralized training produced stronger predictive performance than federated training. The federated model nevertheless retained stage-related discriminatory signal while patient-level training records remained distributed across three simulated hospitals. This supports the feasibility of decentralized genomic model training, but not a predictive advantage over centralized training. The federated configuration was limited to three communication rounds, and convergence was not demonstrated. Any future clinical use would require external and prospective validation in independent patient cohorts before the model could be considered for decision support.

Four-class Stage I–IV prediction remained substantially more difficult than the Early/Advanced formulation. The recurrent models captured more class-balanced stage information than the CNN, but exact stage discrimination remained limited and no model exceeded the majority-class raw accuracy baseline. These findings motivated the use of the binary formulation for the federated learning analysis, while the four-class results were retained to

document the difficulty of exact stage prediction from the available mutation-centred genomic representation.

4.2. Centralized and Federated Learning Performance

The matched centralized BiLSTM performed better than the federated model under the present experimental setting. It achieved accuracy, F1-score, and ROC-AUC values of 0.7022, 0.7072, and 0.7808, compared with 0.6067, 0.6226, and 0.6756 for the federated model. The paired differences were 0.0955 in accuracy (95% CI: 0.0604–0.1306), 0.0845 in F1-score (95% CI: 0.0525–0.1178), and 0.1052 in ROC-AUC (95% CI: 0.0738–0.1375). The ROC-AUC difference was statistically significant under the correlated DeLong test ($p = 8.64 \times 10^{-11}$). These findings indicate stronger predictive performance under centralized training for the matched comparison.

Federated validation performance improved across the three communication rounds. Weighted validation ROC-AUC increased from 0.6432 in round 1 to 0.6853 in round 3, while validation loss decreased from 0.6938 to 0.6626. However, the validation trend had not levelled off by the third round, so convergence could not be established. Because the federated model was still improving at round 3, the observed gap may partly reflect the limited training budget rather than an inherent limitation of federated learning. Federated learning nevertheless allowed collaborative model training while patient-level training records remained distributed across three simulated hospitals, consistent with recent applications of decentralized learning in healthcare and oncology [17, 18].

Local validation performance also varied across clients. Before FedAvg in round 3, local ROC-AUC ranged from 0.6366 to 0.7186, indicating non-uniform performance across the three hospital partitions. This variation may reflect differences in local data or training dynamics, but client-level distribution shift was not formally quantified and its contribution to the centralized–federated performance gap cannot be isolated. Recent federated learning reviews continue to identify client heterogeneity, privacy, scalability, communication constraints, and real-world deployment as important challenges for healthcare applications [19, 20, 21, 22]. Alternative aggregation strategies were not evaluated in the present study.

The centralized–federated comparison should therefore be interpreted within the limited configuration of three communication rounds and one local epoch per client per round. Communication rounds, local training budgets, and aggregation settings were not systematically varied. Consequently, the statis-

tically significant advantage observed for centralized training applies to the configuration evaluated here and should not be interpreted as a general conclusion that centralized learning will always outperform federated learning.

4.3. Challenges of Four-Class Stage Prediction

Four-class Stage I–IV prediction was substantially more difficult than binary Early/Advanced classification. Stage I represented 45.32% of the test set, giving a majority-class accuracy baseline of 0.4532. None of the models exceeded this value: accuracy was 0.4139 for LSTM, 0.3839 for BiLSTM, and 0.2434 for CNN.

Balanced accuracy provided a more informative view under class imbalance. The majority-class classifier would achieve only 0.25, whereas LSTM, BiLSTM, and CNN achieved 0.4038, 0.3732, and 0.2806, respectively. Thus, the recurrent models showed stronger performance across multiple classes than the CNN, although exact stage discrimination remained limited.

Class-wise results further illustrated this difficulty. The LSTM showed its highest recall for Stage IV, while Stage II and Stage III were identified less consistently. CNN predictions were more heavily concentrated in a small number of stages, with most predictions assigned to Stage I or Stage II, only a few to Stage III, and none to Stage IV. During the first three epochs of CNN training, validation accuracy remained near 0.127 even as validation loss decreased from 1.7721 to 1.4888, indicating that the early reduction in loss did not translate into improved exact stage classification. Recent lung cancer AI studies similarly highlight variation in predictive performance across diagnostic, staging, and prognostic applications, together with the continuing need for robust external and prospective validation before clinical implementation [23, 24, 13, 25].

No obvious implementation error was identified in the four-class CNN. The model used four softmax outputs, sparse categorical cross-entropy, correctly encoded four-class labels, and class weights derived from the training distribution. Its weak performance is therefore reported as a limitation of the present architecture and training configuration rather than as evidence of a verified coding error.

Overall, exact Stage I–IV prediction remained challenging. The recurrent models captured more class-balanced signal than the CNN, but performance was not sufficiently strong for reliable exact stage discrimination. These findings motivated the use of the better-performing Early/Advanced binary

formulation for the federated learning analysis, while retaining the four-class results to document the difficulty of more specific LUAD stage prediction.

4.4. Limitations

Some limitations should be considered when interpreting these findings. The dataset included LUAD patients but no matched non-cancer controls, so the framework addresses stage stratification rather than cancer detection. Federated learning was evaluated using three simulated hospital partitions rather than independent clinical institutions, and training was limited to three communication rounds, so convergence was not established. The matched centralized–federated comparison was supported by paired bootstrap confidence intervals and a correlated DeLong test, although the initial BiLSTM checkpoint configuration could not be fully reconstructed from the retained records. The held-out test set had also been evaluated once before BiLSTM continuation training, although it was not used for gradient updates, hyperparameter tuning, stopping decisions, or checkpoint selection. Four-class Stage I–IV prediction also remained limited, with no model exceeding the Stage I majority-class raw accuracy baseline. Transformer-based genomic models such as DNABERT were not evaluated because of the limited data available, which favoured the lighter CNN, LSTM, and BiLSTM architectures.

4.5. Future Work

Future work should include matched non-cancer genomic samples and independent clinical validation to assess discrimination between non-cancer and LUAD cases and improve generalizability [26, 27]. Federated experiments should be extended to more communication rounds, larger numbers of participating clients, different local training budgets, and alternative aggregation strategies. Statistical uncertainty and model comparisons should also be examined in larger independent cohorts to determine whether the performance differences observed in the present study remain consistent across external populations. Larger and better balanced stage groups, additional clinically relevant variables, and improved genomic representations may strengthen exact stage prediction. With larger datasets, future work will move to transformer-based genomic models such as DNABERT for mutation-centred sequence learning. Secure aggregation and differential privacy should also be evaluated in future federated implementations [28, 29]. Finally, prospective work could examine integration with Oxford Nanopore Technologies MinION

and independent clinical cohorts for real-time or near-real-time genomic analysis [30].

5. Conclusion

Accurate LUAD stage stratification from mutation-derived genomic sequences remains challenging because of tumour heterogeneity and the complexity of identifying reliable stage-related genomic patterns. In this study, the corrected 301-bp mutation-centred representation was associated with an increase in CNN ROC-AUC from 0.6832 for the preliminary 101-bp benchmark to 0.7652. For the Early/Advanced classification task, centralized models achieved ROC-AUC values of 0.77–0.78. The matched centralized BiLSTM achieved a ROC-AUC of 0.7808 compared with 0.6756 for the federated BiLSTM, while Stage I–IV prediction remained substantially more difficult and no model exceeded the majority class baseline in raw accuracy. These findings support careful transcript-aware and mutation-centred genomic reconstruction for LUAD stage stratification and show that stage-related predictive signal can also be learned through decentralized training without pooling patient-level training records across the three simulated hospitals. However, federated learning did not provide a predictive performance advantage over centralized training in the present experiment. Overall, this study provides a foundation for mutation-aware LUAD stage stratification and demonstrates the feasibility of decentralized genomic learning, while broader validation is still needed before clinical translation.

Declarations

Conflict of interest. The authors declare that they have no conflicts of interest related to this work.

Ethics statement. This study involved secondary computational analysis of publicly available, de-identified genomic and clinical data. No new human participants were recruited, no biological samples were collected, and no clinical intervention was performed. Ethical approval and informed consent for the original data collection were governed by the respective contributing institutions and source studies. The present work involved only secondary analysis of de-identified data obtained from publicly accessible repositories.

Data availability. The genomic and clinical data used in this study were obtained from publicly accessible LUAD cohorts available through cBioPor-

tal, as described in the Materials and Methods section. The processed 301-bp and preliminary 101-bp mutation-centred datasets, model code, analysis scripts, and federated-learning implementation generated during this study are available from the corresponding author upon reasonable request, subject to the data-use requirements and access conditions of the original repositories.

Code availability. The source code and implementation are publicly available at: **https://github.com/partnerships2024/Federated-Learning-LUAD-Stage-Stratification.git**

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, OpenAI was used to assist with improving the readability and language of the manuscript.The tool was not used to generate experimental results or make scientific decisions. All scientific content, experimental procedures, data analysis, results, and interpretations were reviewed and verified by the authors. Following the use of this tool, the manuscript was reviewed and edited as necessary, and the authors take full responsibility for the content of the publication.

Acknowledgments

The authors gratefully acknowledge the Computational Healthcare Intelligence (CHI) Lab, International Council for Research & Innovation in STE (ICRI-STE), Global Research Community, for providing research support and a collaborative environment that contributed to the development of this study.

References

[1] P. Goldstraw, K. Chansky, J. Crowley, R. Rami-Porta, H. Asamura, W. E. E. Eberhardt, A. G. Nicholson, P. Groome, A. Mitchell, V. Bolejack, The iaslc lung cancer staging project: Proposals for revision of the tnm stage groupings in the forthcoming eighth edition of the tnm classification for lung cancer, Journal of Thoracic Oncology 11 (1) (2016) 39–51. doi:10.1016/j.jtho.2015.09.009.

[2] K. L. Howe, P. Achuthan, J. Allen, J. Allen, J. Alvarez-Jarreta, M. R. Amode, et al., Ensembl 2021, Nucleic Acids Research 49 (D1) (2021) D884–D891. doi:10.1093/nar/gkaa942.

[3] J. T. den Dunnen, R. Dalgleish, D. R. Maglott, R. K. Hart, M. S. Greenblatt, J. McGowan-Jordan, A.-F. Roux, T. Smith, S. E. Antonarakis, P. E. M. Taschner, HGVS recommendations for the description of sequence variants: 2016 update, Human Mutation 37 (6) (2016) 564–569. doi:10.1002/humu.22981.

[4] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, in: International Conference on Learning Representations, 2015.

[5] H. He, E. A. Garcia, Learning from imbalanced data, IEEE Transactions on Knowledge and Data Engineering 21 (9) (2009) 1263–1284. doi:10.1109/TKDE.2008.239.

[6] N. Rieke, J. Hancox, W. Li, F. Milletarì, H. R. Roth, S. Albarqouni, S. Bakas, M. N. Galtier, B. A. Landman, K. Maier-Hein, et al., The future of digital health with federated learning, npj Digital Medicine 3 (2020) 119. doi:10.1038/s41746-020-00323-1.

[7] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, B. A. y Arcas, Communication-efficient learning of deep networks from decentralized data, in: Proceedings of the 20th International Conference on Artificial Intelligence and Statistics, Vol. 54 of Proceedings of Machine Learning Research, 2017, pp. 1273–1282.

[8] P. Kairouz, H. B. McMahan, B. Avent, A. Bellet, M. Bennis, A. N. Bhagoji, et al., Advances and open problems in federated learning, Foundations and Trends in Machine Learning 14 (1–2) (2021) 1–210. doi:10.1561/2200000083.

[9] G. A. Kaissis, M. R. Makowski, D. Rückert, R. F. Braren, Secure, privacy-preserving and federated machine learning in medical imaging, Nature Machine Intelligence 2 (2020) 305–311. doi:10.1038/s42256-020-0186-1.

[10] T. Fawcett, An introduction to roc analysis, Pattern Recognition Letters 27 (8) (2006) 861–874. doi:10.1016/j.patrec.2005.10.010.

[11] T. Saito, M. Rehmsmeier, The precision-recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets, PLOS ONE 10 (3) (2015) e0118432. doi:10.1371/journal.pone.0118432.

[12] A. Fuster-Matanzo, A. Picó-Peris, F. Bellvís-Bataller, A. Jimenez-Pastor, G. J. Weiss, L. Martí-Bonmatí, A. L. Sánchez, D. Bazaga, G. L. Banna, A. Addeo, C. Camps, L. M. Seijo, Á. Alberich-Bayarri, Prediction of oncogene mutation status in non-small cell lung cancer: a systematic review and meta-analysis with a special focus on artificial intelligence-based methods, European Radiology 36 (2026) 2157–2185. doi:10.1007/s00330-025-11962-x.

[13] M. Eftekharian, Z. Hashemi, Artificial intelligence for lung cancer: a systematic review of head-to-head ct, fdg pet/ct, and multimodal models across screening, staging, and prognosis, BMC Medical Imaging (2026). doi:10.1186/s12880-026-02222-5.

[14] E. Zhu, A. Muneer, J. Zhang, Y. Xia, X. Li, C. Zhou, J. V. Heymach, J. Wu, X. Le, et al., Progress and challenges of artificial intelligence in lung cancer clinical translation, npj Precision Oncology 9 (2025) 210. doi:10.1038/s41698-025-00986-7.

[15] N. Liu, G. Han, Q. Gu, Y. Zhang, M. Chen, A new era of precision diagnosis and treatment for lung cancer: artificial intelligence-driven multimodal data integration and clinical applications, Cell Death & Disease 17 (2026) 534. doi:10.1038/s41419-026-08769-z.

[16] T. Chakrabarti, A. G. Mansour, X. Wu, J. Arias-Romero, I. Mambetsariev, N. Chang, S. Delos Santos, T. Mirzapoiazova, J. Fricke, J. Kim, M. Afkhami, C. Lall, A. M. Khan, A. Reyes, M. Lee, D. S. Bruno, C. Ladbury, A. Amini, R. Salgia, Multimodal artificial intelligence in lung cancer: From data integration to precision oncology, Cancers 18 (18) (2026) 2953. doi:10.3390/cancers18182953.

[17] A. Ankolekar, S. Boie, M. Abdollahyan, E. Gadaleta, S. A. Hasheminasab, G. Yang, C. Beauville, N. Dikaios, G. A. Kastis, M. Bussmann, C. Chelala, S. Khalid, H. Kruger, P. Lambin, G. Papanastasiou, OPTIMA Consortium, Advancing breast, lung and prostate cancer research

with federated learning: a systematic review, npj Digital Medicine 8 (2025) 314. doi:10.1038/s41746-025-01591-5.

[18] M. Li, P. Xu, J. Hu, Z. Tang, G. Yang, From challenges and pitfalls to recommendations and opportunities: Implementing federated learning in healthcare, Medical Image Analysis 101 (2025) 103497. doi:10.1016/j.media.2025.103497.

[19] D. Ghosh, M. Mehjabin, M. E. Rayed, M. F. Mridha, M. M. Kabir, Advancements and challenges of federated learning in medical imaging: a systematic literature review, Artificial Intelligence Review 59 (2026) 87. doi:10.1007/s10462-025-11489-z.

[20] S. Bakas, X. Li, P. Shah, H. R. Roth, Federated learning in healthcare: From research to real-world deployment, Annual Review of Biomedical Engineering 28 (2026) 163–186. doi:10.1146/annurev-bioeng-080125-041414.

[21] A. Miloudi, A. Laouid, A. Bounceur, M. Kara, M. M. Bouhamed, et al., Federated learning in healthcare: Recent progress and challenges, Computers & Electrical Engineering 131 (2026) 110924. doi:10.1016/j.compeleceng.2025.110924.

[22] N. Abtahi, C. U. Maimona, M. S. Kabir, Federated learning in healthcare: A comprehensive survey on privacy, scalability and clinical applications, ICT Express 12 (4) (2026) 825–839. doi:10.1016/j.icte.2026.05.011.

[23] X. Yuan, H. Xu, J. Zhu, Z. Yang, B. Pan, L. Wu, H. Chen, et al., Systematic review and meta-analysis of artificial intelligence for image-based lung cancer classification and prognostic evaluation, npj Precision Oncology 9 (2025) 300. doi:10.1038/s41698-025-01095-1.

[24] H. R. Howard, M. Hasanova, A. Tiwari, A. Ghose, R. Winayak, T. Nahar, V. Chauhan, S. Arun, K. Palmer, A. Houston, N. S. Kumar, S. Lalwani, A. Januszewski, J. Conibear, A. Jain, A. Cortellini, A. Passaro, A. Addeo, V. Noronha, G. L. Banna, S. Boussios, The landscape of conventional and artificial intelligence-based clinical prediction models in non-small-cell lung cancer: from development to real-world validation, ESMO Open 10 (9) (2025) 105557. doi:10.1016/j.esmoop.2025.105557.

[25] E.-A. Bonci, A. Bandura, A. Dooley, A. Erjan, H. W. Gebreslase, M. Hategan, D. Khanduja, E. Lai, A. Lescaie, G. V. Nitescu, S. Ramalho, A. Thiha, L. Bertolaccini, Artificial intelligence in nsclc management for revolutionizing diagnosis, prognosis, and treatment optimization: A systematic review, Critical Reviews in Oncology/Hematology 216 (2025) 104929. doi:10.1016/j.critrevonc.2025.104929.

[26] G. S. Collins, J. B. Reitsma, D. G. Altman, K. G. M. Moons, Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): The TRIPOD statement, Journal of Clinical Epidemiology 68 (2) (2015) 134–143. doi:10.1016/j.jclinepi.2014.11.010.

[27] R. F. Wolff, K. G. M. Moons, R. D. Riley, P. F. Whiting, M. Westwood, G. S. Collins, J. B. Reitsma, J. Kleijnen, S. Mallett, PROBAST: A tool to assess the risk of bias and applicability of prediction model studies, Annals of Internal Medicine 170 (1) (2019) 51–58. doi:10.7326/M18-1376.

[28] K. Bonawitz, V. Ivanov, B. Kreuter, A. Marcedone, H. B. McMahan, S. Patel, D. Ramage, A. Segal, K. Seth, Practical secure aggregation for privacy-preserving machine learning, in: Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security, 2017, pp. 1175–1191. doi:10.1145/3133956.3133982.

[29] C. Dwork, A. Roth, The algorithmic foundations of differential privacy, Foundations and Trends in Theoretical Computer Science 9 (3–4) (2014) 211–407. doi:10.1561/0400000042.

[30] M. Jain, S. Koren, K. H. Miga, J. Quick, A. C. Rand, T. A. Sasani, J. R. Tyson, A. D. Beggs, A. T. Dilthey, I. T. Fiddes, et al., Nanopore sequencing and assembly of a human genome with ultra-long reads, Nature Biotechnology 36 (4) (2018) 338–345. doi:10.1038/nbt.4060.